

ConSiDERS-The-Human Evaluation Framework: Rethinking Human Evaluation for Generative Large Language Models

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Abstract

In this position paper, we argue that human evaluation of generative large language models (LLMs) should be a multidisciplinary undertaking that draws upon insights from disciplines such as user experience research and human behavioral psychology to ensure that the experimental design and results are reliable. The conclusions from these evaluations, thus, must consider factors such as usability, aesthetics, and cognitive biases. We highlight how cognitive biases can conflate fluent information and truthfulness, and how cognitive uncertainty affects the reliability of rating scores such as Likert. Furthermore, the evaluation should differentiate the capabilities and weaknesses of increasingly powerful large language models – which requires effective test sets. The scalability of human evaluation is also crucial to wider adoption. Hence, to design an effective human evaluation system in the age of generative NLP, we propose the **ConSiDERS-The-Human** evaluation framework consisting of 6 pillars – Consistency, Scoring Criteria, Differentiating, User Experience, Responsible, and Scalability.

1 Introduction

Generative tasks in natural language processing (NLP) have to rely on human evaluation, as the current set of automated metrics does not correlate well with human judgment (Gao and Wan, 2022; Deutsch et al., 2022). Human evaluation tends to be expensive and difficult to repeat or reproduce (Belz et al., 2023, 2020). Even more importantly, an all-too-common scenario tends to be that the evaluation method is fundamentally misaligned with the problem statement (Hämäläinen and Alnajjar, 2021). In the age of generative large language models (LLMs) with increasing capabilities that can generate fluent content to even fool humans (Clark et al., 2021), ensuring that the human evaluation is

set up appropriately to measure the right aspects and reach the right conclusions is crucial.

In this position paper, first, we argue that to design and interpret the results of human evaluation accurately, the evaluation pipeline needs to be **human-centric** in the age of generative AI, accounting for human evaluators and their cognitive biases. The field of user experience (UX) takes into account the emotional states of a user, a.k.a. *how a user feels* (Marques et al., 2021; Hartson and Pyla, 2012). It is a well-known fact in UX that users tend to be heavily influenced by aesthetic aspects, while actual function or usability aspects take a second place when users perceive a system as useful, leading to the notion “*what is beautiful is useful*” (Sonderegger and Sauer, 2010; Tuch et al., 2012; Hamborg et al., 2014). Aesthetics also extends to language. Factors such as fluency can affect the evaluation, outweighing the actual content or substance (Reber, 2011). Studies in human-computer interface (HCI), cognitive, and social psychology have demonstrated that processing fluency – the ease with which information is perceived and processed in the human mind – has a positive effect on evaluation (Preßler et al., 2023; Tsai and Thomas, 2011; Greifeneder and Bless, 2017). Current state-of-the-art (SOTA) LLMs tend to be quite fluent and produce content that is easy to read and understand, and as a result, users can conflate fluency and usefulness. Therefore, we need to closely examine our human evaluation before reaching conclusions such as – *The LLM can perform function <x> similar to or better than a trained professional*. NLP evaluation procedures, therefore, at the very least must delineate style vs. substance.

Secondly, the **effectiveness of the test set** in measuring the capabilities of a model is critical, as ineffective test sets cannot adequately evaluate these models, a common theme that has surfaced in many leader-boards and public data sets (Tedeschi et al., 2023; Elangovan et al., 2021).

Hence, in this position paper, we make the following contributions:

1) We propose a framework, a structure for organizing and contextualizing human evaluation, that can be customized and adapted to specific contexts. Our proposed framework – the **ConSiDERS-The-Human** evaluation framework – has 6 pillars:

The 6 pillars of ConSiDERS-The-Human Evaluation Framework: (See *Checklist* in Appendix A.1 to follow.)

- **Consistency** of human evaluation: The findings of human evaluation must be reliable and generalizable.
- **Scoring Criteria:** The scoring criteria must include both general purpose criteria such as readability, as well as be tailored to fit the goal of the target tasks or domains.
- **Differentiating:** The evaluation test sets must be able to differentiate the various capabilities as well as the weaknesses of generative LLMs.
- **User Experience:** The evaluation must take into account user experience, including their emotions & cognitive biases, when designing experiments and interpreting results.
- **Responsible:** The evaluation needs to account for responsible AI including aspects such as bias, safety, robustness, and privacy capabilities of the model.
- **Scalability:** Human evaluation must be scalable for pragmatic widespread adoption.

2) We make the case for why UX and the psychology of cognitive biases should be at the forefront of human evaluation. In the last 20 years, less than 7% of the papers (only 16 papers) with “human” and “eval” in their title available in ACL Anthology mention user experience-related keywords in either the title or the abstract (see query in Appendix A.2.6).

3) We highlight how neglecting the role of cognitive biases in human evaluation can lead to incorrect inclusions from the study. We, therefore, provide specific recommendations to mitigate the effects of common cognitive biases. We also provide tips to troubleshoot and improve consistency issues in human evaluation.

In the rest of this paper, we introduce the necessary background concepts in Section 2 and explore each of the 6 pillars in detail in Section 3.

2 Background concepts

2.1 Usability

Usability, according to ISO 9241–11:2018, is defined as – “*The extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use*” (Barnum, 2020). Usability testing includes the following five elements or the 5 Es of usability (Niranjanamurthy et al., 2014; Barnum, 2020). **1) Easy to learn:** This aims to address questions such as a) How easy is it for users to complete basic tasks the first time they use the system? b) When users return to the design after a period of not using it, how well is the user able to recollect how to use the system? **2) Efficiency:** How quickly can experienced users accomplish tasks? **3) Effective:** How completely and accurately the work or experience is completed or goals reached? **4) Error tolerant:** How many errors do users make, how critical are these errors, and how easily can they recover from the errors? **5) Engaging / Satisfaction:** How much does the user like using the system? These 5Es are crucial when designing human evaluation solutions to obtain reliable evaluation results.

2.2 UX and HCI

User experience (UX), according to ISO 9241–110:2010 is defined as – “*a person’s perceptions and responses that result from the use and/or anticipated use of a product, system or service*” (Mirmig et al., 2015). UX, thus, expands beyond the concepts of the 5Es of usability to take into account the broader emotions experienced by the users when using the system (Marques et al., 2021; Hartson and Pyla, 2012). In other words, UX takes into account usability as well as the users’ feelings as to how products “*dazzle their senses, touch their hearts and stimulate their minds*” (Marques et al., 2021). HCI is the UX when humans interact with computer systems, including user interfaces and how information is presented on the digital screen. Commercial LLMs facing end users, such as ChatGPT, have dazzled the minds of their users. Users’ emotion, including perceived usability or usefulness, tends to be heavily influenced by aesthetics (Hartson and Pyla, 2012). Aesthetics influences the user heavily initially or in one-off tests, but over time aesthetics plays a much lesser role and usability becomes crucial (Andreas Sonderegger and Sauer, 2012). The field of UX, therefore, attempts

to disambiguate the perceptions that users form as a result of aesthetics, and the need to measure actual function as defined by the 5Es of usability, whilst embracing user emotions.

2.2.1 Measuring UX: Perception vs. Performance

UX feedback can be qualitative such as user interviews or quantitative metrics. Quantitative UX metrics include performance-based metrics such as time to complete a task, the errors the users encounter, and **perception-based self-reported metrics** through rating scale and preferences (Tullis and Albert, 2013). Likert scale is a type of attitude scale, a special case of a rating scale, that measures the degree to which a person agrees/disagrees with a given statement (Taherdoost, 2019). In NLG, performance-based metrics designed to measure the impact of the end system are considered extrinsic evaluations, while intrinsic evaluations attempt to evaluate properties of the NLG text (van der Lee et al., 2019). Measuring aspects such as fluency are intrinsic evaluations, usually measured through a rating scale or preference tests, where the evaluator is asked which model’s output they prefer. Rating scales and preference tests are based on user perception, and therefore subject to cognitive biases.

2.3 Cognitive biases

Cognitive biases are systematic errors in human judgment or aspects that drive irrational behavior (Tversky and Kahneman, 1974; Ellis, 2018). This is usually a result of relying on heuristics to make a decision. There are many types of cognitive biases, the sources can be broadly categorized into **a)** too much information **b)** lack of information or lack of understanding or meaning associated with the information **c)** need to act or make judgments fast **d)** information that is remembered or recalled (Azzopardi, 2021). For example, when a user is presented with a long list (too much information) during information retrieval, to quickly filter out information, the user would simply click on the first link due to position bias (Azzopardi, 2021).

There are several studies on the effects of cognitive biases on information search and retrieval (Lau and Coiera, 2007; White, 2013) and crowdsourcing (Eickhoff, 2018; Santhanam et al., 2020). For instance, White (2013) finds that evaluation of search and retrieval systems is impacted by confirmation bias – people’s unconscious tendency to prefer information that confirms their beliefs and

disregard evidence that refutes it. There are over 180 different types of cognitive biases identified (Azzopardi, 2021; Tversky and Kahneman, 1974), resulting from a range of factors from how questions are framed (Choi and Pak, 2005) to prior beliefs (White, 2013) that attempt to explain the heuristics humans use to make decisions. These heuristics also impact how humans evaluate LLMs.

3 ConSiDERS-The-Human framework

3.1 User Experience

Cognitive biases play a key role in how humans judge or rate the system. Despite this, there is little reporting of the influence of these biases in NLG tasks with human evaluation (Schoch et al., 2020). In this section, we make the case as to why UX and the psychology of cognitive biases are crucial components of human evaluation in NLP. Since UX and the psychology of cognitive biases are entire fields on their own, it is impossible to cover all the details in this paper. We highlight the significant impact of cognitive biases on human evaluation of NLG tasks that can lead to incorrect conclusions.

Processing fluency – the ease with which information is processed by the human mind (Tsai and Thomas, 2011) – affects factors such as perceived truthfulness and usefulness of statements. These lessons from psychology also apply to NLG human evaluation, where the human evaluation strategy needs to isolate the effects of linguistic fluency vs. aspects such as factual correctness. As presented in the introduction section, the notion “*what is beautiful is useful*” (Sonderegger and Sauer, 2010; Hamborg et al., 2014) also extends to language, where information that is presented in an easy-to-process manner can be perceived as true (Schwarz, 2006). To be clear, we are not calling for linguistic fluency and coherence to be trivialized. On the contrary, we highlight how powerful its influence is on human judgment and evaluation. In the following section 3.1.1, we provide a few tips on how to isolate such effects in NLG evaluation.

3.1.1 Recommendations to mitigate common cognitive biases in NLG evaluation

1. Cognitive uncertainty in user feedback including rating schemes: Explicit user feedback such as 1-5 rating scales, and preference-based tests are inherently subject to cognitive uncertainty, therefore the same user can change their rating on the same item when asked again at a later point in

time, even within a few minutes after the initial rating (Jasberg and Sizov, 2020; Kotkov et al., 2022; Amatriain et al., 2009a). Uncertainty in user feedback is a well-known problem in recommendation systems, where a user’s rating is considered to be noisy (Hill et al., 1995). Jasberg and Sizov (2020) demonstrate the scale of the problem where 65% of the users change their rating even within short intervals between re-rated items. Intra-user rating consistency tends to be higher when the ratings are extreme, e.g., very good or very bad, whereas the in-between ratings tend to have lower consistency (Amatriain et al., 2009a,b). Hence, cognitive uncertainty is more likely to affect model evaluations where the outputs are neither very good nor very bad, where the ratings fall in the mid-point, e.g., 3 on a rating scale of 1 to 5. Similar problems can also surface in preference tests, where users choose “A is marginally better than B” when they don’t have strong opinions.

Mitigation: Rating denoising algorithms in recommendation systems obtain multiple ratings from the same user and attempt to keep only some of the ratings (Joorabloo et al., 2022; Amatriain et al., 2009b). For instance, Amatriain et al. (2009b) propose to keep only those intra-user ratings whose difference is less than a predefined threshold and choose the mildest rating (most neutral rating) as the final rating for that user. The intuition here is that the mildest rating is not likely to affect the items recommended to the user. NLP human evaluation can potentially leverage such denoising algorithms. Some studies have reported that binary preference-based evaluation has *better consistency* compared to rating (Belz and Kow, 2010).

2. Conflate fluency with attributes such as truthfulness: In psychology, the subjective ease with which the mind processes information is more likely to be judged as true (Koch and Forgas, 2012; Reber and Schwarz, 1999). This subjective ease with which the mind processes information can be due to factors including how information is presented, how frequently it is repeated and cues of familiarity, such as native speakers can seem truer than those with a foreign accent (Brashier and Marsh, 2020). Thus, humans rely on shortcuts and draw inferences on aspects such as truthfulness from feelings. Conflating fluency with truthfulness is a result of a cognitive bias called the halo effect. The halo effect is the influence of a global evaluation on evaluations of individual attributes (Nisbett and Wilson, 1977). This particularly af-

fects NLP scoring criteria such as factual completeness, salience, and truthfulness, where these individual attributes can be impacted by the overall global attribute – linguistically fluent easy to process information. Therefore, the reliability of rating schemes particularly affects tasks that require rigorous detailed inspection, such as truthfulness or factual completeness. This cognitive bias has been largely ignored when using rating scales including Likert to evaluate the factual correctness of LLMs. As a result, experimenters can inadvertently conclude that LLMs are as **factually comprehensive** in performing a given function as a trained professional like a doctor. We find that 9 out of the 19 papers we sampled from top-tier medical journals use the perception-based Likert scale to evaluate factual correctness and completeness (details in Appendix A.4), indicating how widespread these practices are.

Mitigation: Tasks such as fact-checking and factual completeness that require inspecting individual traits must be ideally split into atomic facts as detailed in Section 3.2 to isolate the impact of fluency and ease of information processing vs. facts.

3. Over-reliance on initial information: Anchoring bias or over-reliance on an initial piece of information (Tversky and Kahneman, 1974; Furnham and Boo, 2011) affects how models are scored using preference or rating tests. For instance, when performing preference tests, if the model presented first on the screen is always the same, then the initial perception of evaluators can have a significant impact on the rest of the evaluation.

Mitigation: It is important to shuffle the display order so that the order, such as the sequence of human evaluation tasks, doesn’t give away the model. In preference tests, within each task where the outputs of say 2 models (e.g., model A vs. B) are compared, the underlying model representing A and B must also be randomly shuffled so that A does not always refer to the same model.

4. Perception vs. Performance Most self-reported feedback such as user ratings tends to be based on **human perception** (Tullis and Albert, 2013). While perception-based metrics represent how a user feels are necessary to measure subjective aspects such as readability, they do not capture functional performance-based metrics such as efficiency. Studies have shown that users can dislike a system that performs well or like a system that does not perform well (Bailey, 1993). For instance, in an experiment where participants are asked to

choose between one-level and two-level menus for sorting categories, participants preferred the two level menu, even though during actual use the one level menu was much faster and less error-prone (Bailey, 1993; Hayhoe, 1990). This demonstrates the discrepancy between preferences reported vs. measurements of efficiency.

In the context of LLM evaluation, consider a hypothetical application scenario where an LLM generated output is used to automatically draft email responses. In this scenario, how much a user prefers the output of a model versus the reality of how useful that model output is in boosting productivity (time spent drafting emails) need not be correlated. The main challenge with conducting usability studies is that the software system needs to be built to study how it impacts an end user's productivity. In addition, confounding factors such as poorly designed UI can result in reducing productivity and these aspects need to be taken into account when designing and analyzing usability studies to understand the impact of an LLM generated output. Liebling et al. (2022) highlight a similar problem in evaluating large-scale machine translation, where the end user's experience can be quite different from simply evaluating the model output.

Mitigation: NLP tasks that are meant to assist end users, must eventually conduct usability studies to capture performance metrics such as efficiency.

3.2 Consistency of human evaluation

Reproducibility of experiments in science is a widespread challenge and has even led to the term “replication crisis” being coined (Baker, 2016). Human evaluation is no exception to this challenge, where less than 5% of human evaluations are repeatable (Belz et al., 2023). Despite the challenges, consistency cannot be ignored, as poor reproducibility can point to core design problems.

Broadly speaking, **non-random or systematic inconsistencies** primarily arise due to 5 main design flaws **1)** ill-defined evaluation guidelines provided to the annotators **2)** high complexity task **3)** evaluators who are not well qualified or suited to the task **4)** small number of evaluators and/or evaluation set size **5)** rating scales such as the Likert. We specifically need to be able to differentiate between random and non-random inconsistencies, as human evaluators are subject to decision errors/outcomes depending on their cognitive state. Random errors can neither be predicted nor controlled (Sukumar and Metoyer, 2018). Thus, understanding the role

of non-random variations due to system design is key to improving the consistency of evaluation.

1) Ill-defined or complex evaluation guidelines: Ill-defined guidelines are often ambiguous, incomplete, do not address boundary cases, and do not provide adequate examples (Gadiraju et al., 2017). To illustrate this point, Pradhan et al. (2022) use the example of a seemingly simple task “*Is there a dog in this image?*” where the authors point out how even this simple task can elicit several clarification questions such as “Does the dog need to be a real animal?”, “What if the dog is only partially visible in the image?” and “What about a wolf?”. The authors further suggest a 3-stage workflow to improve annotation guidelines: Stage 1 involves workers identifying ambiguous samples; Stage 2 involves labeling a few ambiguous examples to add as clarifying examples in the instructions; and Stage 3 involves workers performing the actual annotation using the revised guidelines with the clarifying examples. Overly long annotation guidelines might even require training the annotators, and hence annotators with task-relevant experience tend to be referred (Rottger et al., 2022). Wu and Quinn (2017) find that using simple vocabulary and logical ordering of instructions can improve the guideline quality. Improving guidelines alone may not be sufficient, as enhancing the user interface design can help reduce cognitive load of the annotator, which in-turn can improve the accuracy of annotation tasks (Alagarai Sampath et al., 2014).

A simple way to identify deficiencies in guidelines is to have “experts” independently evaluate a set of task items using the guideline and compute the IRA score using an appropriate metric such as a Kappa score. Low IRA can indicate potential problems with the annotation guidelines, in which case revising the guidelines iteratively can lead to reasonable agreement (Iskender et al., 2021).

The 5Es of usability testing criteria listed in Section 2.1 is an important strategy to follow when designing human evaluation solutions, including aspects such as how quickly human evaluators can complete their tasks while minimizing errors, how easily the annotation guidelines can be followed and memorized. For example, an overly detailed hard-to-remember evaluation guideline can simply result in poor usability affecting ease of learning, efficiency, and error tolerance resulting in poor inter-rater agreement and/or very slow evaluation turnaround times.

2) High task complexity: Tasks that involve

high cognitive load for the evaluator, can lead to lower agreement (Kim and Park, 2023; Pommeranz et al., 2012; Liu et al., 2023). High cognitive load can even simply involve asking the evaluator to assign a rating of 1-5 (Freitag et al., 2021). One way of mitigating this is to simplify the task. Simplification is key to obtaining consistent results.

An example of task simplification to evaluate the factual completeness or saliency of a model-generated summary is to break a long text into hierarchical units of facts (Liu et al., 2023) using protocols such as Pyramid (Nenkova and Passonneau, 2004). Breaking a reference summary into atomic facts allows the evaluators to verify if the fact is present, and the total number of facts captured in the summary can be summed up automatically to compute the overall score. Thus, the fact-level recall score is likely to yield much more consistent and reliable results compared to a grading scheme, such as the Likert scale, which would involve asking the evaluator to rate the completeness, framing the problem as “*How complete do you think the summary is on a scale of 1-5*”. A flip side to breaking long text into atomic units is that it can lead to loss of information when measuring certain types of criteria, e.g., qualitative aspects such as coherence cannot be evaluated using atomic facts.

3) Ill-suited evaluators: Ill-trained annotators can also be a source of inconsistency, usually a scenario encountered when using crowdsourced workers to evaluate specialized tasks. Annotators who demonstrate poor attention can be identified using a set of attention check questions (Agley et al., 2022). Similarly, ill-trained or ill-qualified workers can be identified using an “exam set”, a test set for which answers are known. Lower IRA can also be a result of variation in the skills or qualifications of the evaluators (Artstein, 2017).

4) Small number of evaluators and/or test set: Low number of participants, or the sample size of the evaluators, is one of the key contributors to poor reproducibility (Maxwell et al., 2015; Button et al., 2013). Using a larger pool of evaluators can mitigate experimenter bias introduced when using very small groups (Sukumar and Metoyer, 2018). A small sample size of the test set is another source that introduces replication problems. A caveat here is that a large group of evaluators and a large test set is necessary but not sufficient to ensure that the study is experiment bias-free, as selection bias can result in a large size that is not representative of the target population (Kaplan et al., 2014).

5) Rating scales such as the Likert: As discussed in Section 3.1.1, self-reported user feedback using rating schemes is inherently noisy and, therefore, unreliable. Despite 50% of human evaluations relying on Likert scale (van der Lee et al., 2021), there has been little investigation into the controversies surrounding it in NLP. These include Likert’s consistency issues (Leung, 2011), aggregation & interpretation of scores (Bishop and Herron, 2015; Willits et al., 2016) and the methods to compute IRA (O’Neill, 2017) that the NLP community needs to research further.

3.2.1 IRA: Importance and caveats

IRA or inter-rater agreement in human evaluation measures how well two or more evaluators agree on the scores or preferences they assign independently. While it is important to measure IRA to detect problems in the design, especially given that only 18% of the papers using human evaluation report IRA (Amidei et al., 2019), there are several caveats called out on the use of IRA metrics in the medical community which we will be discussing below.

Gisev et al. (2013) guide when to use which IRA measure, e.g., Krippendorff’s- α vs. Cohen’s- κ , depending on experiment design factors such as the number of annotators and the type of variable (e.g., ordinal, nominal, etc). Despite such high-level guidance, which IRA measures to use and how to interpret it is contentious (ten Hove et al., 2018; McHugh, 2012). Using an inappropriate metric, such as Fleiss- κ for interval data, is common in NLG (Amidei et al., 2019). Furthermore, even when an appropriate class of IRA metric is used, depending on the IRA chosen, the scores can range from poor to almost perfect (ten Hove et al., 2018).

An intuitive measure of inter-rater agreement is using percent agreement. However, the main criticism was that this does not take chance agreement into account (McHugh, 2012). The question to ask when using IRA metrics, such as a *kappa* statistic is why and when does chance agreement matter? Some tasks genuinely have a class imbalance, e.g., span annotation tasks for named entity recognition, and unmarked spans for the majority class, which would lead to inflated percentage agreement (Artstein, 2017). Another reason is the assumption that some annotators might be making random guesses when they don’t know the answer, and that the majority of the raters may NOT be making deliberate choices (McHugh, 2012). Hence, various IRA measures of Kappa (e.g., Cohen’s- κ) or Alpha (e.g.,

Krippendorff’s- α), estimate the *observed* chance agreement or disagreement empirically when computing IRA. However, if we consider a group of conscientious raters, does chance agreement matter for rating model outputs? For instance, if a model is a high performer and the most common rating is a 4, should disagreeing on a minority rating such as a 1 vs. 2 drastically reduce the IRA? We further demonstrate, using a toy example in Appendix A.3, how disagreement on minority labels can substantially reduce IRA measured using a Krippendorff’s- α while percentage agreement barely changes.

What is considered as “low” IRA can vary from task to task, as complex tasks or difficult samples tend to have low IRA (Kim and Park, 2023). This is crucially important for interpreting IRA and is particularly relevant for NLG evaluation. NLG tasks have typically reported relatively low IRA, e.g., average Krippendorff’s- α of 0.62 (Amidei et al., 2019), the standard interpretation is that experiments with scores less than 0.67 must be deemed unreliable (Marzi et al., 2024). For instance, rating “How good is the generated story?” is more likely to have a lower IRA compared to “Is 99 the largest 2-digit number”.

Hence, while it is mandatory to measure IRA, it is important to ensure that the scores are interpreted in the context of the task. Performing detailed analysis including computing IRA using multiple metrics such as baseline percentage agreement and visualizing the item-wise agreement scores can help analyze the results. Researchers have also called for further investigation to understand the usefulness of a measure for a given problem, demonstrating the challenges in selecting an appropriate IRA measure (ten Hove et al., 2018).

3.3 Scoring criteria

Scoring criteria refers to “What aspects to score”. The 4 common scoring criteria covered in NLP literature are (a) Linguistic Fluency - the quality of single sentence (b) Coherence - overall flow or readability (c) Relevance - importance of content (d) Factuality - factual correctness (Fabbri et al., 2021; Gao and Wan, 2022). Also note that the nomenclature used to indicate a given criterion need not be consistent, e.g., as fluency vs. naturalness, across various studies (van der Lee et al., 2019). Evaluation criteria also need to be customized across NLP tasks, and this is necessary as (a) the criteria will vary between domains and tasks (b) the

generated text almost always needs to be evaluated against multiple criteria (Burchardt, 2013; Freitag et al., 2022). For instance, Freitag et al. (2022) propose the use of Multidimensional quality metrics (MQM) for evaluating machine translation. MQM is a generic framework for evaluating translation quality and provides a catalog of over 100 issues or error types organized in a hierarchy that evaluators can check for (Burchardt, 2013). This hierarchical categorization of errors enables granular as well as coarse-grained analysis of the quality of translation, and can be adapted for NLG evaluations.

In addition to the exhaustive evaluation categorization provided in the MQM framework, responsible AI (RAI) must be factored into human evaluation. Sun et al. (2024) propose 6 categories for RAI evaluation – truthfulness, safety, fairness, robustness, privacy, and machine ethics. Domain-specific customization and extensions also form an integral part of evaluation. For instance, evaluating the effectiveness of an LLM for a domain such as legal should include additional criteria such as case analysis and charge damages calculation (Fei et al., 2023). See conceptual view in Table 1.

High level category	Sub criteria
Core NLP	Fluency
Core NLP	Factuality
Core NLP	Relevance
Core NLP	Coherence
Domain Specific	...
Responsible AI	Bias & Fairness
Responsible AI	Privacy
Responsible AI	Safety
Responsible AI	Robustness

Table 1: Logical view of high-level scoring criteria

3.4 Differentiating

“To differentiate is to identify the differences between things” (vocabulary.com). The test sets used in evaluation must be able to differentiate between the various capabilities as well as the weaknesses of generative LLMs. For instance, when the test sets are relatively easy, most models can achieve high scores and seem very capable. Conversely, when the test sets are very difficult, models might achieve low scores, making the models seem ineffective. Therefore, tests that are ineffective in differentiating capabilities can lead to (a) incorrect conclusions about the capabilities of a model, (b) poor calibration or ranking of various models, or (c) inability to identify any improvements or degradation between model versions, despite spending

significant resources retraining new models. Hence, constructing effective tests that can differentiate model capabilities is crucial, otherwise evaluation simply results in wasted effort.

Traditionally, models have been evaluated using public datasets and benchmark such as GLUE (Wang et al., 2019b) and SuperGLUE (Wang et al., 2019a). These evaluations are not without their share of problems as models can exploit weaknesses in the official test sets relying on shortcuts or spurious correlation – such as length of the input – to predict the target label achieving high performance yet non-generalizable beyond the official test set (McCoy et al., 2019; Elangovan et al., 2023; Gururangan et al., 2018). In addition to these problems, evaluating the current generation of SOTA LLMs poses further challenges. **1)** LLMs are trained on billions of tokens available on the internet, and the training data used is rarely well documented. Hence, these LLMs may have consumed public benchmark test sets as part of their training data (Magar and Schwartz, 2022; Sainz et al., 2023). **2)** The models are generative, producing natural language output unlike a model trained on a classification task, making it difficult to automate evaluation and thus are far more reliant on human evaluation. **3)** The LLMs are capable of following natural language instructions to solve a vast variety of tasks, hence they need to be evaluated on a range of instructions provided as prompts that emulate end-user use cases.

Emulating end-user use cases is crucial, as there are fundamental differences in the tasks in NLP benchmarks vs. the kind of questions or problems users can ask an LLM. Firstly, end users tend to be very creative and prompt the LLM for all kinds of queries. This diversity is particularly important in safety critical applications such as Medicine, as evaluation blindspots can potentially lead to harmful consequences. Secondly, the same message or question can be framed (prompted) in many ways. Hence, test scenarios need to consider the various natural ways in which users write their intentions when interacting with a LLM. If these aspects are not taken account when constructing a test set, solely relying on traditional NLP tasks such as sentiment analysis can result in traditional SOTA models outperforming ChatGPT (Kocoń et al., 2023), when clearly this is not reflective of the end-user experience. Hence, we argue that the test cases used for evaluating LLMs need to represent end user scenarios, in addition to standard NLP tasks.

Benchmarks such as Big-Bench (bench authors, 2023) take a step towards this direction by enabling GitHub contributors to add new tasks to the benchmark. Furthermore, the same prompts may not be effective against all LLMs. Hence, prompts might often have to be customized for individual models, making benchmarking non-trivial.

In light of these new challenges, **curating effective test sets is a critical problem** that the NLP community *must* tackle. Benchmarks such as HELM (Liang et al., 2023) to evaluate LLMs use multiple public datasets across various tasks such as Question Answering (QA) and sentiment analysis and measure aspects beyond accuracy to include toxicity and bias. While these are steps in the right direction, the effectiveness of public test sets such as IMDB dataset (Maas et al., 2011) for sentiment analysis or XSum for summarization (Narayan et al., 2018) in measuring the capabilities of SOTA LLMs can be limited for the reasons discussed earlier. These datasets simply do not sufficiently represent end-user use cases. XSum also has hallucinated content in over 75% of its gold summaries (Maynez et al., 2020), demonstrating further weaknesses in the test sets themselves.

Robustness testing is also a key aspect of evaluation, as models can be vulnerable to basic perturbations such as capitalization, white spaces and prompt formatting (Sclar et al., 2024) and need not understand basic linguistic concepts such as negation (Rogers, 2021). While efforts such as DynaBench to curate progressively harder test sets (Kiela et al., 2021) and behavior testing of models (Ribeiro et al., 2020) are promising approaches to curate effective test sets, curation still needs more research to make significant progress to target end-user use cases to evaluate LLMs.

3.5 Responsible

In human evaluation, the Responsible pillar has to consider two aspects, **a)** Is the model behavior responsible? **b)** Do the human evaluators introduce bias to the evaluation results? While there is no formal definition of Responsible AI, at the very least it entails fairness, safety, truthfulness, and privacy (Sun et al., 2024). Searching papers with “responsible” in their title results in 20 papers, expanding the search to include terms such as “bias” and “privacy” results in ≈ 1000 or 1% of the papers in ACL Anthology, while none of the human evaluation papers mention responsible AI related keywords in their title or abstract (query in Appendix A.2.8), a

telltale sign that more work is needed.

1. Is the model behavior responsible? Answering this question requires evaluating the model for bias, its ability to withstand privacy attacks, truthfulness, and whether the responses are safe. **Bias** is “*prejudice in favor of or against one thing, person, or group*”. Bias affects groups by gender, race, culture, religion, geography, and disability (Esiobu et al., 2023). Thus, the tests sent for human evaluation need to specifically cater to these cases and report the performance of various subgroups to ensure that the LLM behaves responsibly. This requires that the tests have metadata curated such as race to be able to report across these segments.

Generative LLMs are susceptible to leaking private details such as person identifiable information (PII) from the training data, including when subject to **privacy** attacks (Vakili and Dalianis, 2023; Li et al., 2023). One method to mitigate such privacy leaks is to obfuscate PII information such as names and locations from training data using techniques such as Pseudonymization – recognizing privacy-sensitive information and replacing them with realistic substitute (Vakili and Dalianis, 2023) and differential-privacy-based approaches that add noise to the input (Chen et al., 2023). **Safety** ensures that the models do not generate content that harms a person’s physical safety or mental health (Mei et al., 2023; Rusert et al., 2022). Red teams, a group of people authorized to imitate an adversary’s attack or exploitation capabilities, are used to evaluate the robustness, safety, and privacy of LLMs (Perez et al., 2022; Radharapu et al., 2023).

2. Do the human evaluators introduce bias to the evaluation results? As discussed previously, when a substantial portion of evaluation relies on human perception, the diversity of the human evaluators plays a key role in how representative the results are of the wider population. Despite this, less than 3% of papers report demographic information about their evaluators (van der Lee et al., 2021). Having a small group of evaluators or even when the size is large, aspects such as selection bias can result in biased results. Hence, we call for human evaluation to consider the evaluator demographic to mitigate bias effects in the evaluation.

3.6 Scalability

Human evaluation is expensive, yet large volumes of test cases are necessary to differentiate model capabilities and ensure consistency. Hence, optimizations to reduce cost and time is crucial for

wider adoption. Automating parts of human evaluation can reduce cost. For instance, automation might be potentially helpful to shortlist a set of candidates for human evaluation. While the shortcomings of n-gram-based automated evaluations using metrics like Rouge (Lin, 2004) are well studied (Deutsch et al., 2022), approaches such as using LLMs to evaluate LLMs (Lin and Chen, 2023; Chiang and Lee, 2023) need further exploration. Firstly, the effectiveness of LLM-based evaluation is measured using its correlation with human evaluation, hence the human evaluation procedures need to be strengthened first to draw comprehensive and robust conclusions, creating a chicken-and-egg problem. Secondly, LLMs to evaluate LLMs should take into account the differences between perception-based metrics and evaluations that rely on facts, as discussed in Section 3.1.1.

Studies that attempt to reduce the turn-around times of human evaluation itself are limited, less than 50 papers mention “cost” or “scale” with human eval in the title (see Appendix A.2.7 for search query) in the last 20 years. Levinboim et al. (2021) report using coarse-grained caption (as opposed to fine-grained) annotations from crowdsourced users to be able to scale, a method that can be adopted for human evaluation. Huang et al. (2023) propose to identify effective test samples to reduce the cost of human evaluation in conversation systems. Designing a UI that enables the human annotators to work efficiently (one of the Es Efficiency in usability in section 3.1) can reduce time. Given the limited amount of work in this area of scalability of human evaluation, we call for further research.

4 Conclusion & Moving forward

We presented the *ConSiDERS-The-Human* evaluation framework to keep up with the increasing capabilities of LLMs. We highlight the effects of human emotions and cognitive biases on evaluation, given how commonly the perception-based metrics are used to evaluate aspects such as truthfulness. We, hence, encourage researchers in NLP to collaborate with their counterparts in UX, HCI & psychology to ensure that the evaluation *measures the right things the right way* and the results are interpreted accurately. We also call for further research in critical areas – including curating effective test sets, scalability of human evaluation, and responsible AI components such as privacy, bias, robustness, and safety considerations, when evaluating increasingly powerful & ubiquitous generative LLMs.

5 Limitations

Human evaluation is a challenge, especially given the increasing capabilities of SOTA LLMs. Firstly, LLMs have many potential applications and effectively evaluating each application or domain might need customization. Our main aim in this paper is to provide a generic framework extensible for specific domains or applications. Effectively customizing for individual cases might require trial and error. Secondly, perception, whilst important, cannot solely dictate how the quality of LLMs is measured. Humans use heuristics to make decisions, and evaluation has to cater to these heuristics. While there are over 180 cognitive biases, in our paper we only highlight a few that can impact evaluation, we specify this limitation in Section 2.3 as well.

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A Appendix

A.1 ConSiDERS-The-Human Evaluation Checklist

Consistency

Ill-defined or complex evaluation guidelines

1. Were expert annotators independently able to follow the annotation guidelines and achieve higher inter-annotator agreement?
2. Is there a mechanism for evaluators to report issues such as ambiguity in the guideline?

High task complexity

3. Can you simplify the task by breaking down the task into easier tasks?
4. Are you asking multiple questions in a single task? If yes, split each question into separate tasks so that **a)** the answer to one question doesn't bias another one, **b)** the annotators feel positively motivated that they can complete a given task fast.

Ill-suited evaluators

5. Is there a qualification exam for evaluators?
6. Do you insert exam and attention check quality control metrics during evaluation to identify the quality of individual annotators as part of each batch of evaluation?
7. Does your task need fresh annotators? For instance, if your task is to measure how good the LLM's instructions are, you need to make sure that the annotators are new and not used to how to complete the task.

Small number of evaluators and/or test set

8. Is it possible to include more evaluators?
9. Is it possible to increase the size of the test set?

Rating scales such as the Likert

10. Are you trying to measure perception or non-perception-based aspects such as truthfulness? If you are measuring non-perception-based metrics, avoid the Likert scoring, and modify the task to collect many more objective metrics such as extracting facts and verifying each fact.

Inter-rater agreement

11. Which primary IRA measure did you use?
12. Does the primary measure take into account how the evaluation is designed, such as the aspects defined by [Gisev et al. \(2013\)](#)?
13. Do you expect your task to be inherently unbalanced? If not, do you report baseline percentage agreement to verify if the observed chance estimation lowers the overall IRA?
14. Is the item-wise IRA higher for some items and not the others? The ones with lower IRA might be pointing to higher complexity tasks. Report on the distribution of items IRA to troubleshoot the problem.
15. Are the qualifications of the human evaluators similar? If not, the disparity in the evaluators' skills can lead to lower IRA.
16. Do you continuously measure IRA for each evaluation? If yes, did you observe a sudden change in IRA? It might be due to changes in annotators (e.g., adding new annotators) or changes in guidelines or tasks. New guidelines and tasks take a few iterations to settle.

Scoring Criteria

1. Is the model evaluated on typical dimensions: Fluency, Coherence, Relevance, Factuality?
2. What multi-dimensional domain-specific criteria were the model evaluated on?
3. What are the responsible AI criteria the model was evaluated on?

Differentiating

1. How many test cases (test examples) did you use to evaluate each of the criteria?
2. What were the end user use cases, e.g., legal document summarization, were tested? If so, report the end user test cases, their corresponding number of tests, and the scores per user case per criteria.
3. Do you suspect that some of the test cases may have already been used during LLM training? If not sure, answer not sure. If yes, where possible, report the percentage of test cases that may have been impacted.
4. Did you run robustness tests to evaluate model weaknesses? What were the scenarios covered, e.g., semantic preserving perturbations - such as sensitivity to white spaces, synonyms, etc?

User Experience

1. Were rating denoising algorithms applied on the rating-based metrics, such as the Likert, to account for the cognitive uncertainty of the human evaluators?
2. Did you split the content into atomic facts for criteria, such as factual completeness and truthfulness, that require inspecting individual traits of the model-generated text? If so, detail the criteria.
3. Is the Likert scale used appropriately to measure perception-based aspects such as readability?
4. During human evaluation, were examples shuffled properly so that the evaluator cannot tell which model generated which example?
5. Did you test the actual usability of the model's output, in the context of the end-user application (aka extrinsic evaluation)? If so, briefly describe the details.

Responsible

1. Was safety testing performed? If yes, how many test cases were used and what were the test scenarios?
2. Was privacy testing performed? If yes, how many test cases were used and what were the test scenarios?
3. Was the model tested for bias? If yes, what subgroups such as gender or disability was the model evaluated for?

4. How many evaluators were used? What was the diversity of evaluators or several evaluators grouped by aspects such as education qualifications, race, religion, gender, age groups, and nationalities?

Scalability

1. Were parts of the human evaluation automated? If yes, which aspects did you attempt to automate, and what types of automated metrics were used?
2. If you used LLM-based automation, please provide the details of the LLM and prompts.
3. Did you optimize the usability aspects for the annotators to reduce annotation time? If so, provide a summary of how this was achieved.

A.2 Meta-analysis of papers available in ACL Anthology

A.2.1 Papers with “human” and “eval” in either abstract or title

We search for abstracts or titles containing keywords “human” and “eval”, with ≈ 3900 papers, where 900 of those published in 2023 as detailed in Figure 1.

```
(title|abstract=human and title|abstract=eval)
```

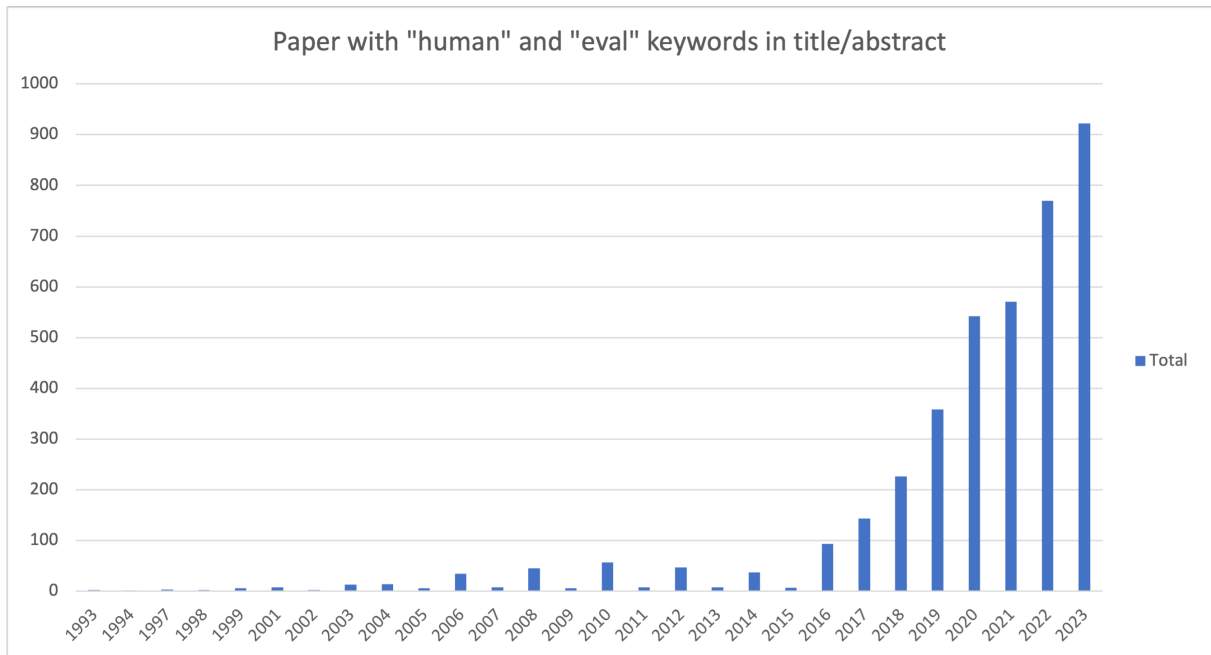


Figure 1: Yearly publications of papers with keywords “human” and “eval” in title/abstract

A.2.2 Papers with “human” and “eval” in the title only

We search for titles containing keywords “human” and “eval”, this results in around 238 papers as shown in Figure 2, reducing the number from 3900 when we include abstracts as seen above.

```
( (title=human and title=eval) )
```

A.2.3 Papers with phrase “human eval” in either abstract or title

We search for abstracts or titles containing phrases human evaluation, and human - evaluation. Around 1300 papers were published.

```
( (title|abstract=[h|H]uman\s?-?\s?[e|E]val) )
```

A.2.4 Papers with keywords “human” and “eval” in title/abstract and usability keywords in either abstract or title

We search for titles or abstracts that mention human evaluation and contain usability keywords (usability, hci, user experience, and human computer). This results in 172 papers. The problem with this query is that it is too noisy as a result of looking for “human” and “eval” in either the title or abstract. For instance,

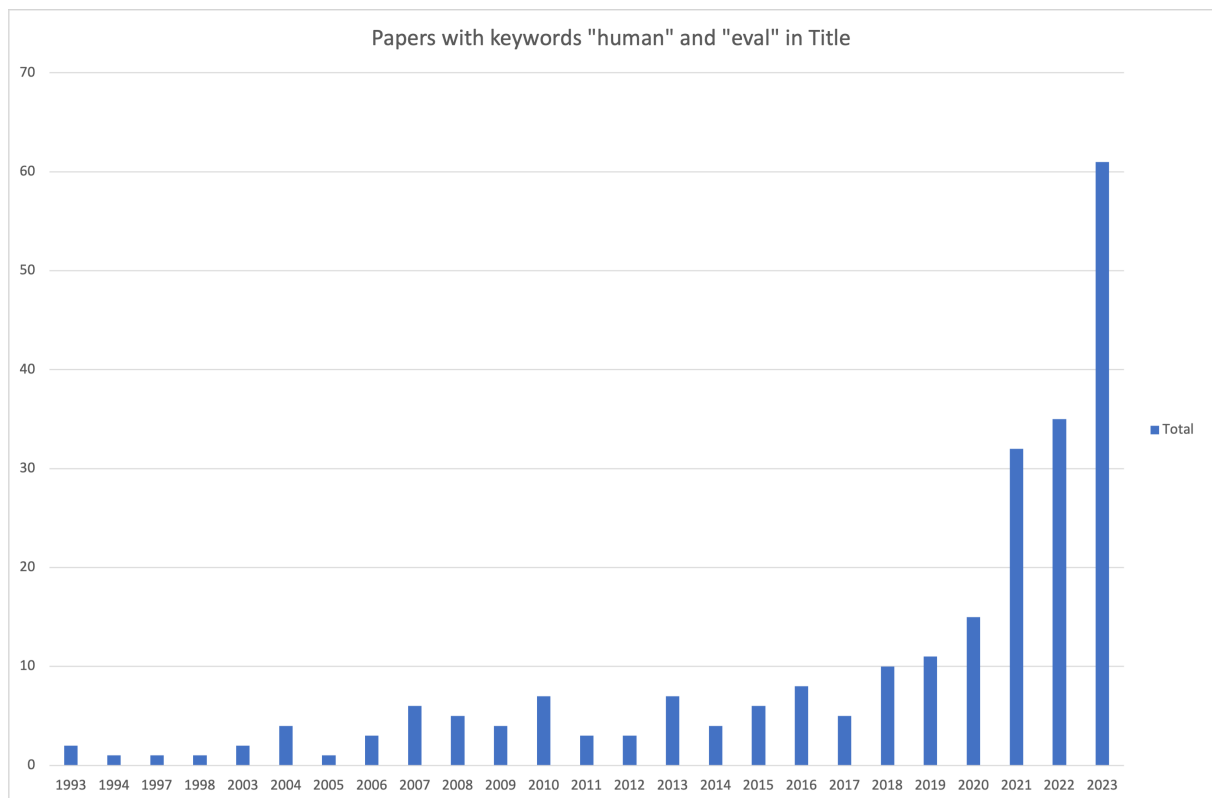


Figure 2: Yearly publications of papers with keywords “human” and “eval” in title

while results contain some human evaluation, the paper’s primary focus is not on the design of human evaluation itself.

Code Listing 1: Jabref search query for human eval and usability

```
(
  (title|abstract=human and title|abstract=eval)
  and
  (
    (title|abstract=usability)
    or (title|abstract=[uU]ser\s[eE]xperience)
    or (title|abstract=user and title|abstract=studies)
    or (title|abstract=hci)
    or (title|abstract=human and title|abstract=computer)
  )
)
```

A.2.5 Papers with phrase “human eval” in either abstract/title and usability keywords

Code Listing 2: Jabref search query for phrase human eval in title and usability in title or abstract

```
(
  (
    (title|abstract=[h|H]uman\s?-?\s?[e|E]val)
  )
  and
  (
    (title|abstract=usability)
    or (title|abstract=[uU]ser\s[eE]xperience)
    or (title|abstract=user and title|abstract=studies)
    or (title|abstract=hci)
    or (title|abstract=human and title|abstract=computer)
  )
)
```

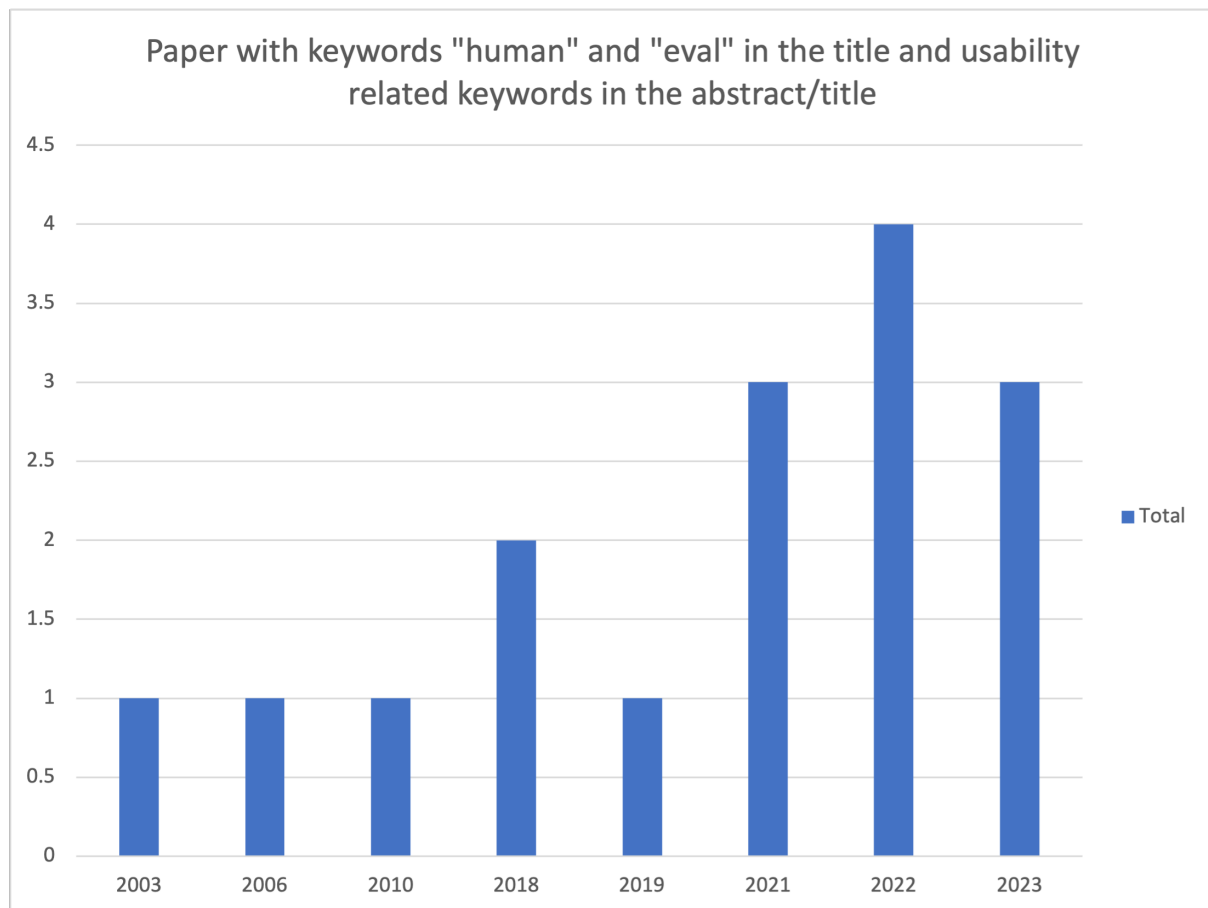



Figure 3: Yearly publications of papers with keywords “human” and “eval” in title and usability keywords in the abstract/title

A.2.6 Papers with keywords “human” and “eval” in title and usability keywords

Out of 238 paper keywords human and eval in the title “(title=human and title=eval)”, only 16 mention usability-related keywords in the title or abstract ACL.

Code Listing 3: Jabref search query for phrase human in title and usability in title or abstract

```
(
  (
    (title=human and title=eval )
  )
  and
  (
    (title|abstract=usability)
    or (title|abstract=user and title|abstract=research )
    or (title|abstract=user and title|abstract=experience )
    or (title|abstract=user and title|abstract=studies)
    or (title|abstract=hci)
    or (title|abstract=human and title|abstract=computer)
  )
)
```

A.2.7 Paper that attempt to scale or optimise human evaluation

Code Listing 4: Jabref search query for phrase human in title and cost / scale in title or abstract

```
(
  (
    (title=human and title=eval )
  )
  and
  (
    (title|abstract=scal)
    or (title|abstract=cost )
    or (title|abstract=time )
  )
)
```

Code Listing 5: Jabref search query for auto eval human in title

```
(
    (title=auto and title=eval )
)
```

A.2.8 Paper that attempt to mention responsible AI related terms in their title

Code Listing 6: Jabref search RAI terms in title

```
(
    (
        (title=responsib)
        or (title=fair)
        or (title=truth)
        or (title=trust)
        or (title=privacy)
        or (title=bias)
        or (title=safe)
    )
    and
    ( not (title=inductive and title=bias))
)
```

Code Listing 7: Jabref search Human eval in title and RAI in asbtract

```
(title=human and title=eval )
and
(
    (
        (abstract|title=responsib)
        or (abstract|title=fair)
        or (abstract|title=truth)
        or (title=trust)
        or (abstract|title=privacy)
        or (abstract|title=bias)
        or (abstract|title=safe)
    )
    and
    ( not (abstract|title=inductive and abstract|title=bias))
)
```

A.3 Krippendorff's Toy example

In this example, we demonstrate with 6 items, 6 raters where each item is rated exactly by 3 raters how simply changing 1 label (row 4, col 2) the Krippendorff's- α drops from 0.7 to 0.24, while percentage agreement only drops from 94.4 to 88.9. In scenario 3, we again just change 1 label, where the % agreement remains the same at 88%, but Krippendorff's- α drops from 0.24 to -0.06.

```
import numpy as np
import krippendorff
from collections import Counter

def compute_agreement(str_reliability_data):
    reliability_data = []

    for coder in str_reliability_data:
        reliability_data.append([np.nan if l == "*" else int(l)
                                for l in coder.split()])

    compute_krippendorff(reliability_data)
    compute_percentage_agreement(reliability_data)

def compute_krippendorff(reliability_data):
    agreement = krippendorff.alpha(reliability_data=reliability_data,
                                    level_of_measurement="nominal")
```

```

print(np.matrix(reliability_data))
print("Krippendorff's alpha for nominal metric: ", round(agreement, 2))

def compute_percentage_agreement(reliability_data):
    """
    Percentage agreement, only for binary values.
    """
    item_wise_data = np.array(reliability_data).T
    item_wise_agreement = []
    print("**Percentage agreement**")
    for item in item_wise_data:
        row_labels = [l for l in item if not np.isnan(l)]
        highest_frequency = Counter(row_labels).most_common(1)[0][1]
        item_percentage = round(100 * highest_frequency / len(row_labels), 2)
        item_wise_agreement.append(item_percentage)

        print(row_labels, item_percentage)

    print("Percentage agreement :", np.mean(item_wise_agreement))

print("***Scenario 1: Baseline ***")
compute_agreement(str_reliability_data=(
    "1    0    *    *    *    *", # coder A
    "1    *    1    1    1    *", # coder B
    "1    0    1    1    1    *", # coder C
    "*    0    1    1    *    1", # coder D
    "*    *    *    *    1    1", # coder E
    "*    *    *    *    *    0", # coder F
))

print()
print("***Scenario 2: Change just 1 label, results in much lower alpha ***")
compute_agreement(str_reliability_data=(
    "1    0    *    *    *    *", # coder A
    "1    *    1    1    1    *", # coder B
    "1    0    1    1    1    *", # coder C
    "*    1    1    1    *    1", # coder D
    "*    *    *    *    1    1", # coder E
    "*    *    *    *    *    0", # coder F
))

print()
print("***Scenario 3: Same % agreement as Scenario 2, but much lower alpha ***")
compute_agreement(str_reliability_data=(
    "1    1    *    *    *    *", # coder A
    "1    *    1    1    1    *", # coder B
    "1    0    1    1    1    *", # coder C
    "*    1    1    1    *    1", # coder D
    "*    *    *    *    1    1", # coder E
    "*    *    *    *    *    0", # coder F
))

```

This produces the following output

```

***Scenario 1: Baseline ***
[[ 1.  0. nan nan nan nan]
 [ 1. nan 1.  1.  1. nan]
 [ 1.  0.  1.  1.  1. nan]
 [nan  0.  1.  1. nan  1.]
 [nan nan nan nan  1.  1.]
 [nan nan nan nan nan  0.]]
Krippendorff's alpha for nominal metric:  0.7
**Percentage agreement**
[1.0, 1.0, 1.0] 100.0
[0.0, 0.0, 0.0] 100.0
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 1.0] 100.0

```

```

[1.0, 1.0, 0.0] 66.67
Percentage agreement : 94.445

***Scenario 2: Change just 1 label, results in much lower alpha ****
[[ 1.  0. nan nan nan nan]
 [ 1. nan 1.  1.  1. nan]
 [ 1.  0.  1.  1.  1. nan]
 [nan 1.  1.  1. nan 1.]
 [nan nan nan nan 1.  1.]
 [nan nan nan nan nan 0.]]
Krippendorff's alpha for nominal metric: 0.24
**Percentage agreement**
[1.0, 1.0, 1.0] 100.0
[0.0, 0.0, 1.0] 66.67
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 0.0] 66.67
Percentage agreement : 88.89

***Scenario 3: Same % agreement as Scenario 2, but much lower alpha ****
[[ 1.  1. nan nan nan nan]
 [ 1. nan 1.  1.  1. nan]
 [ 1.  0.  1.  1.  1. nan]
 [nan 1.  1.  1. nan 1.]
 [nan nan nan nan 1.  1.]
 [nan nan nan nan nan 0.]]
Krippendorff's alpha for nominal metric: -0.06
**Percentage agreement**
[1.0, 1.0, 1.0] 100.0
[1.0, 0.0, 1.0] 66.67
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 1.0] 100.0
[1.0, 1.0, 0.0] 66.67
Percentage agreement : 88.89

```

A.4 Papers in medical journals that use the Likert scale for evaluating ChatGPT

We searched medical journals as follows and manually selected 19 (not cherry-picking) papers that used human evaluation the Likert scale to assess ChatGPT. We find that the Likert scale was used to evaluate factual completeness, saliency correctness in 9/19 papers, 4/19 of papers use the Likert scale appropriately to measure user perception and for the rest of 6/19 we were not sure of what the criteria meant, for details see Table 3.

- **Nature & sub-journals:** Performed search within the nature website using keywords ⟨gpt medical . likert⟩
- **Lancet:** Performed search within the Lancet website using keywords ⟨gpt likert⟩.
- **JMIR:** Searched for ⟨gpt likert, jmir⟩ on Google Scholar and used the first 2 pages of results to filter relevant context.

Title	Relevant quote from paper	Dimensions	Likert used for factual correctness / completeness
<i>Nature & Subjournals</i>			
Harnessing ChatGPT and GPT-4 for evaluating the rheumatology questions of the Spanish access exam to specialized medical training	The medical experts evaluated the clinical reasoning of the chatbots followed in each of the responses. Their evaluation was based on a 1–5 scale, where a score of 5 indicates that the reasoning was entirely correct and flawless, while a score of 1 signifies that the reasoning was inconsistent or contained significant errors.	1) overall correctness.	Yes
A pilot study on the efficacy of GPT-4 in providing orthopedic treatment recommendations from MRI reports	[in Table 2] Likert scales used for (a) Treatment recommendations are clinically useful and relevant (b) Treatment recommendations are based on scientific and clinical evidence (c) The overall quality of the treatment recommendations	1) overall quality; 2) based on evidence; 3) useful and relevant; 4) up-to-date; 5) consistent.	Yes
Testing the limits of natural language models for predicting human language judgements	No details mentioned in the paper	1) overall quality	Not sure
Availability of ChatGPT to provide medical information for patients with kidney cancer	The SERVQUAL model is a research tool that assesses how five dimensions—tangibility, reliability, responsiveness, assurance, and empathy—influence customer perception. The answers to the questions are presented in a five-point Likert scale. SERVQUAL has mainly been used to evaluate the quality of medical services in hospitals and healthcare institutions.	1) tangibility; 2) reliability; 3) responsiveness; 4) assurance; 5) empathy.	Yes
Explaining machine learning models with interactive natural language conversations using TalkToModel	We evaluated the following statements along the 1–7 Likert scale at the end of the survey: Easiness: I found the conversational interface easier to use than the dashboard interface; Confidence: I was more confident in my answers using the conversational interface than the dashboard interface; Speed: I felt that I was able to more rapidly arrive at an answer using the conversational interface than the dashboard interface; Likelihood to use: based on my experience so far with both interfaces, I would be more likely to use the conversational interface than the dashboard interface in the future.	1) Easiness; 2) Confidence; 3) Speed; 4) Likelihood to use.	No
Evaluating large language models on medical evidence summarization	We systematically evaluate the quality of generated summaries via human evaluation. We propose to evaluate summary quality along several dimensions: (1) Factual consistency; (2) Medical harmfulness; (3) Comprehensiveness; and (4) Coherence. These dimensions have been previously identified and serve as essential factors in evaluating the overall quality of generated summaries. ... The order in which the summaries are presented is randomized to minimize potential order effects during the evaluation process. We utilize a 5-point Likert scale for the evaluation of each dimension.	1) Factual consistency; 2) Medical harmfulness; 3) Comprehensiveness; 4) Coherence.	Yes
A large-scale comparison of human-written versus ChatGPT-generated essays	The questionnaire covers the seven categories relevant for essay assessment shown below: Topic and completeness; Logic and composition; Expressiveness and comprehensiveness; Language mastery; Complexity; Vocabulary and text linking; Language constructs. ... These categories are used as guidelines for essay assessment 44 established by the Ministry for Education of Lower Saxony, Germany. For each criterion, a seven-point Likert scale with scores from zero to six is defined, where zero is the worst score (e.g. no relation to the topic) and six is the best score (e.g. addressed the topic to a special degree). The questionnaire included a written description as guidance for the scoring.	1) Topic and completeness; 2) Logic and composition; 3) Expressiveness and comprehensiveness; 4) Language mastery; 5) Complexity; 6) Vocabulary and text linking; 7) Language constructs.	Yes
People devalue generative AI's competence but not its advice in addressing societal and personal challenges	Participants were asked to rate the author competence on three items: The author is knowledgeable of the subject; The text is credible; I intend to follow the provided recommendations.	1) knowledgeable 2) credible 3) willing to follow.	Not sure
Quality of information and appropriateness of ChatGPT outputs for urology patients	The responses generated by ChatGPT were then compared to those provided by a board-certified urologist who was blinded to ChatGPT's responses and graded on a 5-point Likert scale based on accuracy, comprehensiveness, and clarity as criterias for appropriateness.	1) accuracy; 2) comprehensiveness; 3) clarity.	Yes

Table 2: Papers containing “GPT”, “medical” and “Likert” from Nature and Subjournals.

Title	Relevant quote from paper	Dimensions	Likert used for factual correctness / completeness
<i>Lancet</i>			
Assessing the potential of GPT-4 to perpetuate racial and gender biases in health care: a model evaluation study	Factual correctness and humanness of letters were assessed by two independent clinicians using a Likert scale ranging from 0 to 10, with 0 representing completely incorrect or inhuman and 10 representing completely correct and human.	two dimensions: correctness and humanness.	Yes
<i>Journal of Medical Internet Research (JMIR)</i>			
Putting ChatGPT's Medical Advice to the (Turing) Test: Survey Study	Participants were also asked about their trust in chatbots' functions in patient-provider communication, using a Likert scale from 1-5. ... On average, responses toward patients' trust in chatbots' functions were weakly positive (mean Likert score 3.4 out of 5), with lower trust as the health-related complexity of the task in the questions increased.	1) trustworthy	Not sure
A Generative Pretrained Transformer (GPT)-Powered Chatbot as a Simulated Patient to Practice History Taking: Prospective, Mixed Methods Study	To assess how our participants perceived the simulated patient, we used the Chatbot Usability Questionnaire (CUQ). This 16-item questionnaire measures the personality, user experience, error management, and onboarding of a chatbot and has recently been validated. ... For the CUQ, we provided relative numbers of Likert categories.	1) Personality; 2) User Experience; 3) Error Handling; 4) Onboarding; 5) Other	Yes
Health Care Trainees' and Professionals' Perceptions of ChatGPT in Improving Medical Knowledge Training: Rapid Survey Study	The questionnaire was designed according to the Kirkpatrick model, with four dimensions to understand the thoughts of the students: (1) perceived knowledge acquisition (KA), (2) perceived training motivation (TM), (3) perceived training effectiveness (TE), and (4) perceived training satisfaction (TS). Three experts reviewed and edited the questionnaire, which has 12 questions, including 1 open-ended question. A 5-point Likert scale was adopted for all questionnaire items (from 1=strongly disagree to 5=strongly agree).	1) perceived knowledge acquisition (KA); 2) perceived training motivation (TM); 3) perceived training effectiveness (TE); 4) perceived training satisfaction (TS).	No
Assessing Health Students' Attitudes and Usage of ChatGPT in Jordan: Validation Study	The survey tool was created based on the TAM framework. It comprised 13 items for participants who heard of ChatGPT but did not use it and 23 items for participants who used ChatGPT. ... Each item was evaluated on a 5-point Likert scale with the following responses: strongly agree scored as 5, agree scored as 4, neutral/no opinion scored as 3, disagree scored as 2, and strongly disagree scored as 1. The scoring was reversed for the items implying a negative attitude toward ChatGPT.	13 items	No
Increasing Realism and Variety of Virtual Patient Dialogues for Prenatal Counseling Education Through a Novel Application of ChatGPT: Exploratory Observational Study	Sentences were then appraised by a neonatologist for realism, relevance, and usability for virtual prenatal counseling simulations. Each metric used a 5-point Likert scale, ranging from 1 (the lowest) to 5 (the highest).	1) realism; 2) relevance; 3) usability.	Not sure
ChatGPT Versus Consultants: Blinded Evaluation on Answering Otorhinolaryngology Case-Based Questions	The questions were answered by both ORL consultants and ChatGPT 3. ORL consultants rated all responses, except their own, on medical adequacy, conciseness, coherence, and comprehensibility using a 6-point Likert scale.	1) medical adequacy; 2) conciseness; 3) coherence; 4) comprehensibility	Not sure
Evaluation of GPT-4's Chest X-Ray Impression Generation: A Reader Study on Performance and Perception	In a blind randomized reading, 4 radiologists rated the impressions based on "coherence", "factual consistency", "comprehensiveness", and "medical harmfulness", which were used to generate a radiological score based on a 5-point Likert scale of each dimension.	1) coherence; 2)factual consistency; 3) comprehensiveness; 4) medical harmfulness.	Yes
Investigating the Impact of User Trust on the Adoption and Use of ChatGPT: Survey Analysis	We developed 2 latent constructs based on the question (predictors): Trust and Intent to Use. Participant responses to all the questions were captured using a 4-point Likert scale ranging from 1=strongly disagree to 4=strongly agree. The Actual Use factor, the outcome variable, was captured using a single-item question capturing the frequency of use ranging from 1=once a month to 4=almost every day.	1) trust; 2) intent to use.	Not sure
Exploring the Possible Use of AI Chatbots in Public Health Education: Feasibility Study	Medical students' feedback was collected anonymously at the end of the training experience through a 3-item questionnaire with a Likert scale (1 to 10) regarding their general satisfaction, willingness to repeat the experience, and ease of use of the tool. In particular, the scale of the 3 items can be translated as follows:; Item 1: 1="dissatisfied with the experience", 10="very satisfied"; Item 2: 1="I would not repeat the experience", 10="I would definitely repeat the experience"; Item 3: 1="the tool is too difficult to be used", 10="the tool was very easy to be used".	1) satisfy; 2) intent to use; 3) easy to use.	No

Table 3: Papers containing "GPT" and "Likert" from Lancet and JMIR.